Introduction to MPI

ECM3446/ECMM461: High Performance Computing

Dr. Man Luo

University of Exeter

m.luo@exeter.ac.uk

Module Overview

Week 1: Introduction to HPC	Week 5: Floating point arithmetic	Week 8: Task parallelism and manger worker parallelism
Week 2: Introduction to OpenMP	Week 6: Memory and cache	Week 9: Linear algebra and matrices
Week 3: Numerical solutions to PDEs	Week 7: Factors affecting parallel performance	Week 10: Accelerators and GPUs
Week 4: Introduction to MPI		Week 11: Summary

Syllabus Plan

- Motivation of and introduction to high-performance computing
- Parallel computer architectures: shared-memory and distributed-memory architectures, multi-core processors, Graphics Processing Units (GPUs)
- Parallel programming methods: OpenMP, Message Passing Interface (MPI)
- Floating point arithmetic: floating point model, range, accuracy, exceptions
- Parallel processing algorithm and programming design: domain decomposition, halo exchange, manager-worker, task-based parallelism
- Parallel performance: speed-up, efficiency, parallel overheads and scaling
- Interconnection networks in high performance computers: topologies, latency and bandwidth

MPI overview

- OpenMP only works when processors share memory
- Can use MPI with both **distributed memory** and **shared memory**
- **MPI enables us to program for distributed memory systems like clusters**
- With MPI we run multiple copies of our program which communicate by passing messages
  - Multiple processes with separate memory address spaces
  - Make calls to functions to send and receive messages
- See SAB Chapter 8 ("The Essential MPI")

MPI libraries and standards

- Functions are provided by an external library with C and Fortran interfaces.
- A number of implementations are available
- Open source:
  - OpenMPI (www.open-mpi.org)
  - MPICH (www.mpich.org)
- Standard is under active development (www.mpi-forum.org)

Lecture Units

- Unit 4.1: MPI basics
- Unit 4.2: Point-to-point communication
- Unit 4.3: Collective communication
- Unit 4.4: Non-blocking communication
- Unit 4.5: Domain decomposition